Journal Pre-proof



LC-HRMS-based target, suspect and non-target screening of Greek peaches and nectarines: pesticide occurrence, transformation products and dietary risk assessment

Vasileios D. Alampanos, Lelouda-Athanasia Koronaiou, Styliani Petromelidou, Anthia D. Gkantou, Theocharis G. Danis, Dimitra A. Lambropoulou

PII: S0308-8146(26)04099-9

DOI: https://doi.org/10.1016/j.foodchem.2026.151939

Reference: FOCH 151939

To appear in: *Food Chemistry*

Received date: 1 July 2026

Revised date: 14 October 2026

Accepted date: 1 November 2026

Please cite this article as: V.D. Alampanos, L.-A. Koronaiou, S. Petromelidou, et al., LC-HRMS-based target, suspect and non-target screening of Greek peaches and nectarines: pesticide occurrence, transformation products and dietary risk assessment, *Food Chemistry* (2026), https://doi.org/10.1016/j.foodchem.2026.151939

This is a PDF of an article that has undergone enhancements after acceptance, such as the addition of a cover page and metadata, and formatting for readability. This version will undergo additional copyediting, typesetting and review before it is published in its final form. As such, this version is no longer the Accepted Manuscript, but it is not yet the definitive Version of Record; we are providing this early version to give early visibility of the article. Please note that Elsevier's sharing policy for the Published Journal Article applies to this version, see: https://www.elsevier.com/about/policies-and-standards/sharing#4-published-journal-article. Please also note that, during the production process, errors may be discovered which could affect the content, and all legal disclaimers that apply to the journal pertain.

© 2026 Published by Elsevier Ltd.

Journal Pre-proof

LC-HRMS-based target, suspect and non-target screening of Greek peaches and nectarines: pesticide occurrence, transformation products and dietary risk assessment

Vasileios D. Alampanos[1,2*], Lelouda-Athanasia Koronaiou[1,2], Styliani Petromelidou[3], Anthia D. Gkantou[3], Theocharis G. Danis[3], Dimitra A. Lambropoulou[1,2]

[1]*Laboratory of Environmental Pollution Control, Department of Chemistry, Aristotle University of Thessaloniki, Thessaloniki, GR-541 24, Greece*

[2]*Centre for Interdisciplinary Research and Innovation (CIRI-AUTH), Balkan Center, 10th km Thessaloniki-Thermi Rd, Thessaloniki, GR-570 01, Greece*

[3]*ANALYSIS - DELCOF S.A., Quality Control and Chemical Analysis Laboratory, Kopanos, Naousa, Greece*

**Corresponding author:*
E-mail address: vdalampa@chem.auth.gr

Journal Pre-proof

Journal Pre-proof

Abstract

Peaches and nectarines (*Prunus persica*) are widely consumed stone fruits requiring comprehensive residue monitoring. Herein, an integrated LC-HRMS workflow combining target, suspect, and non-target screening was applied to 74 samples collected from the two major Greek production subregions. Target analysis enabled the quantification of 22 pesticide residues with cyprodinil, tebuconazole, difenoconazole, boscalid, fludioxonil, etofenprox, and pyraclostrobin among the most frequently detected. Comparison with EU MRLs showed only two slight exceedances, corresponding to 2.7% of samples, below the recent EFSA-reported average. Target-driven suspect screening, supported by literature-derived MS data, retention time prediction, and in silico MS/MS fragmentation, enabled the tentative annotation of nine transformation products associated with five pesticides. Non-target screening expanded chemical characterization by annotating emerging contaminants and endogenous compounds. Dietary risk assessment indicated acute and chronic exposure below toxicological reference values, suggesting no appreciable consumer risk. The workflow demonstrates the value of integrated LC-HRMS for pesticide monitoring and broader chemical profiling.

Journal Pre-proof

Keywords: peaches, nectarines, LC-HRMS screening, transformation products, pesticides, dietary risk assessment, food safety, high resolution mass spectrometry (HRMS)

1. Introduction

The use of pesticides in fruit cultivation plays an important role in protecting crops from pests and diseases, thereby supporting yield stability and helping to maintain uniformity in physicochemical and visual quality characteristics (e.g., fruit size and appearance) critical for market acceptance. (Fenik et al., 2011; *Pesticides | EFSA*, 2026; Tudi et al., 2021) However, the occurrence of pesticide residues in fruits remains a significant food safety concern due to potential chronic dietary exposure and associated health risks for the general population. (Åkesson et al., 2023; Jankowska et al., 2024; Sandoval-Insausti et al., 2022). Peaches and nectarines (Prunus persica L.) are major stone fruits, with global production averaging 23.11 million metric tons annually over 2016–2025 and projected to reach 23.83 million metric tons in 2025/2026, led by China (*FAOSTAT*, 2023; USDA Foreign Agricultural Service, 2026; World Population Review, 2026). Greece contributes approximately 3–4% of global peach production and about 26% of European Union production, ranking as the third-largest producer in the EU after Spain and Italy, and is consistently reported as one of the world's leading exporters of canned peaches, accounting for roughly 55% of global exports in recent years. (Enterprise Greece, 2026; *FAOSTAT*, 2023; ReportLinker, 2023). Within Greece, the regions of Imathia and Pella in Central Macedonia account for more than 80% of the national peach and nectarine production, making them strategically important for both domestic consumption and international trade (*Statistics - ELSTAT*, n.d.).

Their cultivation requires repeated plant-protection interventions against fungal diseases and insect pests, which may result in multiple pesticide residues at harvest, with studies in peaches and other stone fruits reporting the co-occurrence of compounds such as boscalid, bupirimate, difenoconazole, pyraclostrobin, and pyrimethanil. (Bakırcı et al., 2014; EFSA, 2026; Keklik et al., 2025; *Pesticides | EFSA*, 2026; Si et al., 2025; Słowik-Borowiec et al., 2015; Tudi et al., 2021) Botanically, peaches and nectarines belong to the same species, *Prunus*

persica, with nectarines representing a genetic variant characterized by a recessive allele responsible for the absence of fruit skin pubescence (trichomes). The primary morphological difference between the two fruits is therefore the smooth skin of nectarines compared to the fuzzy skin of peaches, while their overall biochemical composition, agronomic practices, and pesticide treatment regimens are similar. Thus, both are mainly considered together in production statistics and residue monitoring programs. (FAO/WHO Codex Alimentarius Commission, n.d.; Keklik et al., 2025; Marion Murray et al., 2020)

Given their high consumption and economic significance, systematic monitoring of pesticide residues in peaches and nectarines is necessary for consumer health, product quality and regulatory compliance. International organizations establish maximum residue limits (MRLs) and promote harmonized risk assessment approaches (FAO/WHO, 2026). In the European Union, pesticide residues are regulated under Regulation (EC) No 396/2005, while active-substance approval is governed by Regulation (EC) No 1107/2009. The European Food Safety Authority (EFSA) supports this framework through annual residue monitoring and dietary risk assessment. (EFSA et al., 2024, 2025, 2026). Comparable regulatory oversight is provided in the United States by the Environmental Protection Agency (US EPA, 2015). These frameworks require robust analytical methods for comprehensive residue monitoring. However, regulatory MRLs primarily address parent active substances, while cumulative exposure to multiple residues remains an important consideration, particularly for vulnerable population groups. Moreover, increasing attention has been directed toward pesticide transformation products (TPs), which may arise from environmental degradation, plant metabolism, or post-harvest processes and may contribute to overall dietary exposure (Alampanos & Lambropoulou, 2025; García-Reyes et al., 2007; C. Li et al., 2021; Mahler et al., 2021; Mitra et al., 2024). Notably, certain TPs have been reported to exhibit enhanced toxicity compared to their parent compounds as for example, fipronil sulfone and desnitro-

imidacloprid (Romero et al., 2016; L. Yang et al., 2020). However, TPs are not included in routine monitoring. Therefore, both parent pesticides and TPs should be monitored for a more realistic consumer risk assessment.

Conventional pesticide monitoring relies predominantly on targeted multi-residue methods based on triple quadrupole mass spectrometry coupled with liquid or gas chromatography, which offer high sensitivity and selectivity for predefined analytes. Nevertheless, such methods are restricted to specific compounds and do not permit retrospective data analysis. (Bakırcı et al., 2014; Eyring et al., 2021; Hernández et al., 2013; Keklik et al., 2025; Słowik-Borowiec et al., 2015) Newly introduced pesticides, TPs, or unexpected contaminants may therefore remain undetected. In this context, LC-HRMS has emerged as a powerful tool for comprehensive food analysis, as full-scan accurate-mass acquisition enables target quantification, suspect screening, and non-target analysis (NTA) within a single analytical run. (Gkountouras et al., 2024; Kaufmann, 2012; Kunzelmann et al., 2018). The identification confidence of compounds detected through suspect and non-target workflows can be reported using the framework proposed by Schymanski et al.(Schymanski et al., 2014), enhancing transparency and reproducibility. A structured and rational approach to investigating pesticide TPs involves constructing suspect lists derived from pesticides detected in the samples and from compounds frequently reported in the literature for peaches and nectarines. This target-driven suspect strategy increases analytical relevance while minimizing false-positive annotations and unnecessary data complexity. Such workflows align with recent recommendations in high-resolution food analysis that advocate integration of regulatory monitoring with discovery-based screening. LC-HRMS full-scan data also provide information on endogenous metabolites, which may reflect cultivar, ripening stage, agronomic practices, and environmental conditions and are therefore widely used in food authenticity studies. However, Imathia and Pella are geographically adjacent and share the same cultivation and

environmental conditions. Thus, endogenous metabolites were not used for geographical classification but to complement pesticide-related findings and broaden the chemical characterization of peach and nectarine matrices. (Cao et al., 2021; Z. Li et al., 2022; Tamasi et al., 2021; Zhao et al., 2022)

Despite considerable advances in pesticide residue analysis, studies integrating targeted pesticide quantification, TPs oriented suspect screening, comprehensive non-target profiling of xenobiotic and endogenous compounds, and dietary risk assessment within a unified LC-HRMS workflow remain limited. Most investigations focus either on regulatory monitoring of pesticide residues or on food metabolomics, rarely combining these complementary analytical strategies. The study was designed on the premise that integrated LC-HRMS analysis would provide broader chemical characterization than target analysis alone. In this context, the present work implemented a tiered LC-HRMS approach for peaches and nectarines originating from the two major Greek production regions, integrating (i) targeted quantification of pesticide residues, (ii) suspect screening of pesticide TPs based on detected parent pesticides and literature information, (iii) non-target screening for the annotation of additional pesticide-related compounds and endogenous metabolites, and (iv) dietary risk assessment based on the detected pesticide residues, providing a comprehensive chemical characterization of peaches and nectarines while demonstrating the added value of full-scan LC-HRMS for both routine pesticide monitoring and broader food chemical profiling.

2. Materials and methods

2.1 Chemicals and reagents

Analytical standards of pesticides included in the target analysis were obtained from certified suppliers (Sigma Aldrich, Dr. Ehrenstorfer) with stated purities higher than 95%. LC-

MS-grade solvents (methanol, acetonitrile and water) were purchased from Thermo Fisher Scientific (Germany). Individual stock solutions (1000 mg L^{-1}) were prepared in acetonitrile and stored in amber screw-capped glass vials in the dark at −20 °C. Working standard mixtures were prepared by appropriate dilution with acetonitrile. LC-MS grade (99%) formic acid (HCOOH) was purchased from Sigma Aldrich and used as a mobile phase acidifier. QuEChERS extraction salts including magnesium sulfate ($MgSO_4$), sodium chloride (NaCl), trisodium citrate dihydrate ($Na_3C_6H_5O_7 \cdot 2H_2O$), and disodium hydrogen citrate sesquihydrate ($C_6H_6Na_2O_7 \cdot 1.5H_2O$) were obtained from Sigma Aldrich. Clean-up sorbents including primary secondary amine (PSA), C18, and anhydrous $MgSO_4$ were also bought from Sigma Aldrich. Prior to LC injection, each sample was filtered through 0.22 m glass fiber syringe filters (Millipore, Cork, Ireland) using sterile disposable syringes. Polypropylene Falcon-type centrifuge tubes (Sarstedt, Nümbrecht, Germany) were used for sample handling and processing.

2.2 Sample collection and storage

A total of 74 peach and nectarine samples were collected from 74 privately owned commercial orchards located in the two major Greek peach-producing regions, Imathia and Pella. Sampling was conducted during June and July, corresponding to the main harvesting period for peaches and nectarines in Northern Greece. Information on the sampling locations, harvest year, regional climatic conditions, and agroproduction practices is provided in Table S1. The exact number of individual orchard owners/producers was not available, as ownership information was not recorded. For each orchard, a representative sample of approximately 2 kg, corresponding to about 8–12 fruits depending on fruit size, was collected using a five-point sampling approach. Specifically, fruits were collected from five spatially distributed locations across each orchard, taking orchard size and layout into account and following agronomic guidance to ensure representative sampling of each production site. Information on the exact

interval between the last pesticide application and sampling was not systematically recorded. After collection, samples were transported to the laboratory in insulated coolers with ice packs at approximately 4 °C and subsequently stored at −20 °C until analysis. Prior to extraction, all fruits from each orchard sample were homogenized together at ambient temperature for 2 min in a single cycle using a commercial 650 W blender (Kenwood) to obtain a representative composite matrix.

2.3 LC-HRMS instrumentation

A Vanquish UHPLC system coupled to a Q Exactive™ Focus Orbitrap high-resolution mass spectrometer (Thermo Fisher Scientific, Bremen, Germany) equipped with a heated electrospray ionization source (HESI-II) was used. The instrument was operated in fast polarity-switching mode. Chromatographic separation was performed on a Hypersil GOLD aQ column (50 × 2.1 mm, 1.9 μm) maintained at 40 °C. The mobile phase consisted of (A) LC–MS water and (B) methanol, both containing 0.1% formic acid. The gradient elution program (Table S2) was applied at an initial flow rate of 0.200 mL min^{-1} as follows: 10% B (0–1.5 min), increased to 60% B at 4.0 min and 70% B at 8.0 min, followed by 100% B at 11.0 min and held until 13.0 min. The system was then returned to 10% B at 14.0 min for column re-equilibration until 15.0 min. Additional LC parameters are summarized in Table S3. High-resolution mass spectrometric analysis was performed using a data-dependent acquisition method (Full MS/ddMS²). Full-scan MS spectra were acquired at a resolving power of 70,000 (FWHM) over an *m/z* range of 100–1000. Data-dependent MS/MS spectra were acquired at a resolving power of 35,000, using an isolation window of 1 *m/z* and stepped normalized collision energies (NCE) of 15, 30, and 50. The AGC target was set to 1×10^6 for MS and 2×10^5 for MS/MS scans, with the maximum injection time set to auto. Up to five precursor ions were selected per scan cycle using an apex trigger of 2–5 s, while isotopic peaks were excluded. Spectra were acquired in centroid mode. Detailed HESI source parameters and additional MS and MS/MS conditions

are provided in Table S4. Instrument control and data acquisition were performed using Xcalibur 4.1 (Thermo Fisher Scientific Inc., Waltham, MA, USA). Data processing, compound identification, and feature extraction were carried out using TraceFinder 4.1 and Compound Discoverer 3.2 (Thermo Fisher Scientific, Bremen, Germany).

2.4 Sample preparation workflow

Sample preparation was performed using a modified citrate-buffered QuEChERS workflow adapted for the extraction and clean-up of pesticide residues from peach and nectarine matrices. Initially, 10 g (±0.1 g) of homogenized sample were transferred into 50 mL polypropylene centrifuge tubes and extracted with 10 mL acetonitrile. The mixture was vigorously shaken for 1 min and subsequently subjected to ultrasonic extraction for 10 min in order to improve analyte partitioning and extraction efficiency. Following extraction, phase separation was induced by the addition of a QuEChERS salt mixture consisting of 4 g $MgSO_4$, 1 g NaCl, 1 g trisodium citrate, and 0.5 g disodium hydrogen citrate sesquihydrate. The tubes were immediately shaken vigorously for 1 min to prevent salt agglomeration and facilitate efficient partitioning between the aqueous and organic phases. Samples were then centrifuged for 10 min at 4000 rpm to obtain separation of the acetonitrile extract from the sample matrix. For extract clean-up, 5 mL of the upper acetonitrile phase were transferred into 15 mL centrifuge tubes containing dispersive solid-phase extraction (d-SPE) sorbents consisting of 50 mg primary secondary amine (PSA), 100 mg C18, and 150 mg $MgSO_4$. The mixture was shaken vigorously for 1 min and centrifuged again for 10 min. Finally, the purified extract was filtered through a 0.22 µm PTFE syringe filter and transferred into LC vials prior to LC-HRMS analysis. Procedural blanks and matrix-spiked samples were included throughout the analytical workflow to monitor potential contamination and extraction performance.

Journal Pre-proof

2.5 Target analysis of pesticide residues

Target screening and quantification of pesticide residues were performed using TraceFinder® 4.1 software (Thermo Fisher Scientific). A comprehensive target list of 285 pesticides (Table S5), previously reported by our research group (Anagnostopoulou et al., 2022), was incorporated into the TraceFinder quantitative processing method. Analytical standards were used to establish retention times, exact masses, and MS/MS fragmentation patterns, which were subsequently used for compound confirmation. Quantification was carried out using matrix-matched calibration curves, prepared by spiking blank peach and nectarine extracts with a multi-residue pesticide standard mixture containing the compounds included in the target list. Method performance was evaluated in terms of linearity, limits of detection (LOD), limits of quantification (LOQ), accuracy (recovery), repeatability (RSD), and matrix effects. LODs and LOQs were estimated from the signal-to-noise ratio of the lowest matrix-matched calibration standard (0.001 mg/kg), using S/N thresholds of 3 and 10, respectively. Recovery and repeatability were assessed from six replicate spiking experiments at 0.010 mg kg^{-1}. Matrix effects were evaluated by comparing the slopes of matrix-matched and solvent-based calibration curves and were expressed as percentage signal suppression or enhancement. Analyte-specific expanded measurement uncertainty was not experimentally determined, as full validation across multiple concentration levels was beyond the scope of the present study. Detailed validation and analytical data of interest for the 22 quantified pesticides included in the target database are provided in Table S7. The remaining compounds in the 285-pesticide target list were included for screening purposes but were not detected and were therefore not quantitatively evaluated in the present study. Positive identification and confirmation of target compounds were performed considering the applicable recommendations of DG SANTE 11312/2021-v2 (European Commission, Directorate-General for Health and Food Safety, 2026), based on retention-time agreement (±0.1 min), precursor-

ion mass accuracy (±5 ppm), isotopic-pattern matching (±5 ppm; fit ≥70%), at least one characteristic MS^2 fragment (±5 ppm), and S/N ≥10. Signals below 1×10^4 intensity were considered background. Detected pesticide concentrations were subsequently compared with the maximum residue limits (MRLs) established under Regulation (EC) No 396/2005, as retrieved from the European Commission EU Pesticides Database (European Commission, n.d.), in order to evaluate compliance with current European regulatory requirements.

2.6 Suspect screening workflow for transformation products

Suspect screening was performed in TraceFinder™ 4.1 (Thermo Fisher Scientific) using the full-scan LC-HRMS data. A target-driven workflow was applied, focusing exclusively on TPs and metabolites associated with the 22 pesticides detected during target analysis. A study-specific suspect database comprising 216 unique TPs and metabolites was compiled from the scientific literature and publicly accessible resources, including PubChem, the NORMAN Suspect List Exchange, Eawag-BBD, and the FAIR-TPs platform. (LCSB, n.d.; Mayahi et al., 2026; Kang et al., 2020; Schusterova et al., 2025; Y. Yang et al., 2025; Kiefer et al., 2019; Man et al., 2021; Chen et al., 2025; PubChem, n.d.; Mohammed Taha et al., 2022) The complete final list is provided in Table S6 of the Supplementary Material. It comprised both experimentally reported TPs and transformation products previously predicted using computational tools such as BioTransformer (Chen et al., 2025). Exact masses, molecular formulae, MS/MS fragmentation data and structural information were collected whenever available. Candidate features were screened using accurate-mass filtering, isotopic pattern matching and comparison of experimental MS/MS spectra against reference fragments available in the database. When experimental fragmentation data were unavailable, in silico fragmentation was performed using CFM-ID (Wang et al., 2022), and the predicted spectra were compared with the experimentally acquired data-dependent MS^2 ($ddMS^2$) spectra. To minimize false-positive annotations, candidate features were filtered using S/N >10, peak area

>2 × 10^6, mass accuracy within ±5 ppm, isotopic fit >90%, acceptable chromatographic peak shape, retention-time consistency (±0.1 min), and agreement with reference or predicted MS/MS fragments. All tentative annotations were subsequently manually inspected using Xcalibur™ 4.1. Only candidates fulfilling both the automated filtering criteria and the manual verification requirements were retained for further evaluation. The resulting candidate list was further refined using retention time prediction. A prediction model was trained on the target pesticides of known retention time under the present chromatographic conditions using Retip (Bonini et al., 2020), and the predicted retention times were used as an additional, supporting criterion to support or reject tentative annotations particularly for candidates lacking confirmatory MS/MS evidence (Figure S1). Final identification confidence levels were assigned based on the combined evidence from accurate-mass measurements, isotopic pattern matching, fragmentation data, retention time information, literature reports and *in silico* predictions, following the framework proposed by Schymanski et al (Schymanski et al., 2014).

2.7 Non-Target Screening Workflow for Emerging Contaminants and Endogenous Compounds

To extend the chemical characterization of peach and nectarine samples beyond target and suspect screening, a non-target analysis workflow was applied to the full-scan LC-HRMS data using Compound Discoverer™ 3.2. A customized workflow based on the Compound Discoverer™ "Food Research w Stats Unknown ID w Online and Local Database Searches" template was implemented, with the individual processing steps and parameters described below and summarized in Figure S2 and Table S8. Raw LC-HRMS data were processed within a retention time window of 0.5-13.5 min using a mass accuracy threshold of 5 ppm throughout the workflow. Retention time alignment was performed using an adaptive curve model, followed by compound detection with a minimum peak intensity of 5 × 10^5 and a signal-to-noise threshold of 10. Detected features were subsequently merged and grouped across samples

using a retention time tolerance of 0.2 min. Compound grouping considered the preferred ions $[M+H]^+$, $[M+Na]^+$ and $[M+NH_4]^+$ in positive ionization mode and $[M-H]^-$ in negative ionization mode. Feature annotation combined elemental composition prediction, mzCloud MS/MS spectral matching, local mass-list screening, ChemSpider database searching, spectral distance evaluation, and mzLogic scoring. Molecular formula prediction was performed using isotope-pattern and fragment-ion matching, while mzCloud searches relied on cosine similarity matching of experimental and library MS/MS spectra. Local database screening was performed against the Varita Lab 6549 Flavonoid Structure Database and the EFS HRAM Compound Database, whereas ChemSpider was searched using exact mass and molecular formula across selected environmental, food, toxicological, and regulatory databases (Table S8). All databases were accessed in October 2025. Candidate annotations were subsequently evaluated using spectral distance matching and mzLogic scoring. Final compound assignments were generated by integrating evidence from mzCloud, mzVault, local mass-list searches, predicted elemental compositions, and ChemSpider results through the Compound Annotation node. Missing values were recovered using the Fill Gaps node, while background features were filtered using procedural blanks. Features exhibiting a sample-to-blank area ratio ≤ 5 were classified as background and excluded from further evaluation. Only compounds assigned to Schymanski Level 2a confidence, based on high-quality MS/MS spectral matching against mzCloud, were considered for reporting and discussion in the present study.

2.8 Dietary risk assessment

To evaluate the potential health risks associated with the dietary intake of the pesticide residues detected in the samples, both acute and chronic dietary exposure were estimated following the methodology recommended by the FAO/WHO Joint Meeting on Pesticide Residues (JMPR) (Cui et al., 2025; FAO/WHO, 2026; World Health Organization, n.d.). The acute risk was assessed through the acute Hazard Quotient (aHQ), derived from the estimated short-term

intake (ESTI) and the acute reference dose (ARfD), while the chronic risk was assessed through the chronic Hazard Quotient (cHQ), derived from the estimated daily intake (EDI) and the acceptable daily intake (ADI), as described in Equations (1)-(4):

$$ESTI = \frac{HR \times LP}{bw} \quad (1)$$

$$aHQ = \frac{ESTI}{ARfD} \quad (2)$$

$$EDI = \frac{C_{mean} \times F}{bw} \quad (3)$$

$$cHQ = \frac{EDI}{ADI} \quad (4)$$

where *HR* is the highest residue concentration measured for a given pesticide (mg/kg), C_{mean} is the mean residue concentration across all samples (mg/kg), *LP* is the large-portion consumption of peaches/nectarines (kg day^{-1}), F is the average daily fruit consumption (kg day^{-1}), and bw is the mean body weight (kg). Residue concentrations below the LOQ were substituted with LOQ/2, and non-detects were assigned a value of zero, in accordance with the lower–middle-bound substitution approach for left-censored monitoring data (EFSA, 2010). For pesticides for which an ARfD was deemed unnecessary by the regulatory authority, the acute assessment was not applicable, and these compounds were excluded from the aHQ calculation. The cumulative dietary risk arising from simultaneous exposure to multiple residues was expressed as the Hazard Index (HI), calculated as the sum of the individual chronic Hazard Quotients of all detected pesticides (Equation 5):

$$HI = \sum_{i=1}^{n} cHQ_{\mathrm{i}} \quad (5)$$

Journal Pre-proof

A pesticide was considered to pose no appreciable risk to consumers when its HQ was below 1, and the cumulative exposure was regarded as acceptable when HI < 1; values ≥ 1 would indicate a potential health concern (Kottadiyil et al., 2023). Toxicological reference values (ADI and ARfD) were obtained from the EU Pesticides Database and the JMPR evaluations. Food consumption inputs (average and large-portion consumption of peaches) and the corresponding mean body weights were retrieved from the EFSA Pesticide Residue Intake Model (PRIMo) rev. 3.1 (EFSA, 2019; European Commission, n.d.), using the most critical (worst-case) European consumer populations identified by the model for this commodity. For the acute assessment, the large-portion values were 446.7 g day^{-1} for the adult (general-population) scenario (body weight 65.8 kg) and 183.6 g day^{-1} for the children scenario (body weight 10.2 kg); for the chronic assessment, the highest average peach consumption among the European dietary surveys was used (0.376 and 0.348 g kg^{-1} bw day^{-1} for the adult and children scenarios, respectively). Both scenarios were evaluated to account for the higher intake-to-body-weight ratio in younger consumers.

3. Results and Discussion

3.1 LC-HRMS target analysis results

3.1.1 Occurrence and detection frequency of targeted pesticide residues

In total, 22 parent pesticides compounds were detected or quantified in the analyzed samples, belonging to two major pesticide classes: fungicides (n = 12) and insecticides (n = 10). Quantification was performed for each compound using matrix-matched calibration curves, which showed good linearity for all 22 quantitatively evaluated pesticides, with R^2 values ranging from 0.9920 for etofenprox to 0.9995 for penconazole (Table S7). Mean

Journal Pre-proof

recoveries at 0.010 mg kg^{-1} ranged from 70.6% to 110.8%, with RSD values of 6.5–14.7%, demonstrating satisfactory trueness and repeatability under the evaluated validation conditions. LODs ranged from 0.00015 to 0.00090 mg kg^{-1}, while LOQs ranged from 0.00050 to 0.00300 mg kg^{-1}. These values lie well below the lowest applicable EU MRL for any monitored compound, ensuring that any regulatory exceedance could be reliably detected and quantified. Regarding matrix effects, signal suppression predominated, affecting 20 of the 22 compounds (−1.9% to −50.0%), with strong suppression (<−20%) observed for etofenprox (−50.0%), τ-fluvalinate (−39.5%), flonicamid (−36.4%), chlorantraniliprole (−22.5%), and acetamiprid (−22.0%). Signal enhancement was observed only for difenoconazole (+12.5%) and tetraconazole (+7.9%). As matrix effects exceeded ±20% for five compounds, matrix-matched calibration was applied to all target pesticides to compensate for signal suppression or enhancement and ensure reliable quantification.

Regarding the analytical results (Table S9), all 74 samples contained at least one residue above the LOD, and 72 samples (97.3%) had at least one residue quantified above the LOQ. Only two samples were free of any quantifiable residue. Across all samples, 260 quantified detection events were recorded. Fungicides dominated the residue profile in terms of both occurrence and quantified detections, accounting for 145 events (55.8%), compared with 115 events (44.2%) attributed to insecticides, including the insect growth regulator pyriproxyfen and the insecticide/acaricide τ-fluvalinate. Fungicide predominance became even more pronounced when sub-LOQ detections were considered, indicating that several fungicides occurred as persistent low-level residues, whereas insecticides were more frequently present at concentrations exceeding LOQ. This pattern is consistent with the preventive and season-long use of fungicides to manage fungal diseases in stone-fruit orchards, in contrast to the more targeted and phenology-driven application of insecticides.

[Figure 1]

Journal Pre-proof

Tebuconazole, emamectin benzoate, boscalid, cyprodinil, and trifloxystrobin were among the most frequently quantified pesticides (Figure 1; Table S9). When sub-LOQ signals were also considered, the detection pattern shifted, with cyprodinil and fludioxonil showing the highest overall occurrence. The difference between detection and quantification frequencies for cyprodinil (95.9% vs 25.7%), fludioxonil (91.9% vs 2.7%) and pyrimethanil (58.1% vs 1.4%) indicates very frequent low-level occurrence of these fungicides in the orchards, consistent with routine orchard fungicide use of combination products (e.g. Switch®, containing cyprodinil and fludioxonil; Scala®, containing pyrimethanil) (Thomidis & Michailides, 2009), but that spray-to-harvest intervals are, in most cases, sufficient to allow residues to dissipate in very low concentrations.

The higher fungicide detection frequency across both fruit types and regions is consistent with the importance of fungal disease management in stone-fruit production across Mediterranean Countries, particularly against *Monilinia spp.* (brown rot), which often requires repeated fungicide applications during the growing season. (Casals et al., 2021). Within the insecticide group, several compounds were of distinctly minor occurrence. The insecticide/acaricide τ-fluvalinate was both detected and quantified in only 2 samples (2.7%), and spinetoram (a spinosyn-class insecticide) in a single sample (1.4%) at 0.017 mg kg^{-1}. The insect growth regulator pyriproxyfen was detected in 8.1% of samples (n = 6) but quantified in only 3 (4.1%), indicating mostly trace-level occurrence. These lower occurrence frequencies may reflect the more targeted, pest- and phenology-dependent use of insecticides compared with the broader preventive use of fungicides.

3.1.2 Residue concentration ranges and multi-residue co-occurrence

Fungicide residue concentrations varied considerably among compounds and samples (Figure 2, Table S9). Boscalid showed the broadest concentration distribution, while tebuconazole was the most frequently quantified fungicide. Despite their high detection

frequencies, cyprodinil and particularly fludioxonil were predominantly present below the LOQ, with fludioxonil quantifiable in only two samples. This pattern suggests substantial residue dissipation before sampling. Previous studies have shown that fludioxonil can undergo photodegradation, which may contribute to its environmental dissipation. (Apell et al., 2019; J. Yang et al., 2020). Pyraclostrobin and trifloxystrobin were also generally present at low concentrations. Taken together, these findings indicate that the frequent occurrence of fungicides across the sampled orchards was predominantly associated with low residue concentrations at harvest. Among insecticides, etofenprox showed the highest maximum concentration and the broadest concentration distribution (Figure 2; Table S9), which may partly reflect its high lipophilicity (log $K_{ow} \approx 6.9$) and potential retention on fruit surfaces. Emamectin benzoate was the most frequently quantified insecticide but was generally present at low concentrations, while acetamiprid and chlorantraniliprole showed intermediate occurrence and concentration levels. The occurrence of chlorantraniliprole is consistent with its use against lepidopteran pests in stone-fruit production, particularly *Grapholita molesta*, in peach and nectarine production. (*Grapholita Molesta - an Overview | ScienceDirect Topics*, n.d.). Spinosyn A and spinosyn D were co-detected in the same nine samples, consistent with their joint presence as the major components of spinosad formulations (Ueno et al., 2006). Their maximum summed concentration remained below the corresponding EU MRL (Table 1). τ-Fluvalinate was only occasionally quantified. Thus, the insecticide residue profile was more heterogeneous, with marked compound-specific differences in both occurrence and concentration levels.

[Figure 2]

Among all quantified residues, boscalid showed the highest single-sample concentration (0.373 mg kg^{-1}), followed by etofenprox (0.310 mg kg^{-1}), tebuconazole (0.139 mg kg^{-1}), cyprodinil (0.132 mg kg^{-1}), acetamiprid (0.084 mg kg^{-1}) and fludioxonil (0.041 mg

kg^{-1}, quantifiable in only 2 samples) (Table S9; Figure 3). The number of quantified residues per sample ranged from 0 to 8, with a mean of 3.51 ± 1.82 (Figure S3). Only two samples (2.7%) were free of any quantified residue. Profiles of 2–5 simultaneously quantified compounds accounted for the majority of samples. Eight compounds were co-quantified in the most heavily contaminated specimen (Sample 68; peach, July, Imathia).

[Figure 3]

To further visualize multi-residue patterns and concentration variability among samples, a heatmap was constructed using residues quantified above the LOQ (Figure 4). The heatmap illustrates differences in pesticide occurrence and concentration across samples, with fungicides generally showing greater occurrence than insecticides. Several compounds also exhibited characteristic co-quantification patterns. In particular, boscalid and pyraclostrobin were frequently quantified in the same samples, consistent with their presence in commercial co-formulations such as Bellis® or Signum®, while spinosyn A and spinosyn D consistently co-occurred, reflecting their joint presence in Spinosad formulations.

[Figure 4]

The cumulative quantified pesticide load was assessed by summing the residue concentrations ($\Sigma_{concentration}$) within each sample (Figure S4). Total residue concentrations varied considerably among samples, ranging from non-quantifiable levels to 0.935 mg kg^{-1}, although most samples showed relatively low cumulative concentrations (median 0.049 mg kg^{-1}). The two most contaminated samples, Sample 68 (0.935 mg kg^{-1}) and Sample 67 (0.682 mg kg^{-1}) were both peaches collected in late July from Imathia and were the only two samples exceeding 0.50 mg kg^{-1}. Fungicides generally accounted for the largest proportion of the total residue load, whereas substantial insecticide contributions were restricted to a smaller subset of samples, primarily due to etofenprox (Figure S4). The marked variability observed among

Journal Pre-proof

samples highlights the heterogeneous nature of pesticide-use practices across the surveyed orchards, even within the same production region and harvesting period, with only a limited number of samples contributing disproportionately to the cumulative residue burden.

3.1.3 Compliance with EU Maximum Residue Limits

All detected residue concentrations were compared against applicable EU MRLs under Regulation (EC) No 396/2005 and its amendments (European Commission, n.d.). The authorization status of the corresponding active substances was also evaluated separately from MRL compliance. 21 compounds were authorized for use on peaches and/or nectarines in Greece at the time of sampling, whereas tricyclazole was not authorized. For 20 compounds, all measured concentrations were below the corresponding MRLs (Table 1). For spinosad, for which the MRL applies to the sum of spinosyn A and D, the maximum summed concentration was also below the applicable MRL.

Measured concentrations numerically above the corresponding MRLs were observed in two samples (2.7%). Acetamiprid was measured at 0.084 mg kg^{-1} in one sample (MRL = 0.080 mg kg^{-1}), representing a marginal numerical exceedance. Tetraconazole (MRL = 0.03 mg kg^{-1}) exceeded the regulatory limit in one sample (Sample 62; peach, Imathia, July), with a measured concentration of 0.056 mg kg^{-1} (1.86-fold exceedance). Tetraconazole is a (triazole) fungicide authorized in Greece for stone fruit (European Commission, n.d.). This exceedance is indicative of a late-season application without adequate pre-harvest interval. The overall exceedance rate of 2.7% (2/74) is consistent with, and slightly below, rates from European monitoring programs (EFSA et al., 2025, 2026). However, as analyte-specific expanded measurement uncertainty was not taken into consideration, these findings should not be considered as fully confirmed regulatory non-compliances.

Tricyclazole merits separate discussion. Historically authorized against rice blast but with EU approval revoked (Commission Implementing Regulation (EU) 2016/1826) since

2017 (European Commission, 2016), it was quantified in five samples (6.8%) at 0.002–0.009 mg kg^{-1}. The applicable default MRL is 0.01 mg kg^{-1} (set at the limit of determination for non-approved substances), and all five findings were below this threshold. However, MRL compliance does not imply authorization of use. Therefore, the occurrence of tricyclazole requires further investigation regarding its origin, including possible unauthorized use or contamination from agricultural inputs.

[Table 1]

3.2 Suspect Screening results: Identification of Transformation Products

Application of the suspect screening workflow enabled the tentative identification of nine TPs associated with five pesticides detected by target analysis: difenoconazole, tebuconazole, cyprodinil, flonicamid, and spinosad. Detailed identification and occurrence data are provided in Table S10. Two TPs were assigned to Schymanski Level 2b and seven to Level 3. TEB_338 and CYP_404 were the most frequently detected TPs, followed by FLO_192 and SPIN_732, whereas the remaining TPs occurred less frequently across the analyzed samples (Figure 5). The distribution of TP signal intensities is further illustrated in Figure S6 using $\log_{10}$-transformed peak areas. Comparison with the corresponding parent pesticides revealed that TPs generally occurred less frequently than their parent compounds (Figure 5). This pattern was particularly evident for cyprodinil, tebuconazole, and difenoconazole, which were widely detected whereas their associated TPs occurred in a smaller proportion of samples. Spinosad represented an exception, with its associated TP showing the same detection frequency as the parent compound. These findings indicate that the occurrence of a parent pesticide did not necessarily coincide with detectable levels of its corresponding TPs.

[Figure 5]

Journal Pre-proof

Parent-only detections predominated across all pesticide groups, while parent–TP co-occurrence was observed for tebuconazole, cyprodinil, flonicamid, spinosad, and difenoconazole. Only one flonicamid TP was detected in the absence of its parent compound, whereas all spinosad TP detections coincided with the parent pesticide. These patterns further demonstrate that TP screening provided complementary information beyond parent-pesticide occurrence. The predominance of parent-only detections may reflect the persistence and relative stability of the parent compounds and incomplete transformation under the conditions represented by the sampled fruits. However, TP occurrence is also influenced by compound-specific formation and degradation kinetics and analytical detectability. Moreover, the toxicological relevance of the detected TPs remains uncertain because compound-specific toxicity data are limited. Further investigation is therefore required to clarify both the occurrence and fate of these TPs and their potential toxicological relevance.

[Figure 6]

3.3 Non-target analysis results for pesticides and endogenous compounds.

The non-target screening workflow generated many features that were annotated and classified according to the framework proposed by Schymanski et al. To ensure high confidence, only compounds assigned to Level 2a were considered for further discussion. Regarding pesticide-related compounds, NTA confirmed the same pesticide residues already identified by target analysis, with no additional high-confidence pesticide annotations retained. Therefore, the non-target results were interpreted mainly in relation to food-contact-related compounds and endogenous metabolites. In addition to pesticide-related compounds, the NTA workflow revealed several other compounds potentially associated with food-contact materials, polymer additives, packaging, or post-harvest handling processes. These compounds were not detected in procedural blanks and were present in a large proportion of the analyzed samples, supporting their relevance as sample-associated features rather than laboratory background

contamination. The identified compounds included plasticizers and polymer additives such as bis(2-ethylhexyl) terephthalate, bis(4-ethylbenzylidene)sorbitol, and bis(methylbenzylidene)-sorbitol, antioxidant-related compounds such as 2,2′-methylenebis(4-methyl-6-tert-butylphenol) and 3,5-di-tert-butyl-4-hydroxybenzaldehyde, as well as other industrial or packaging related compounds including 4-methylbenzophenone, dibenzoylmethane, triphenylphosphine oxide, and 4-dodecylbenzenesulfonic acid (Kunzelmann et al., 2018; PubChem, n.d.). Their frequent occurrence highlights the added value of NTA for revealing non-pesticide chemical residues that may arise from packaging, transport, storage, or broader environmental exposure pathways.

Moreover, the NTA workflow enabled the annotation of 21 endogenous metabolites at confidence Level 2a (Table S11). The identified compounds belonged to several major biochemical classes, including phenolic acids, flavonoids, amino acids, sugars, fatty acids, oxylipins, triterpenoids, and other lipid-derived metabolites, reflecting the chemical complexity of peach and nectarine fruits.(Covarrubias et al., 2020; Lillo-Carmona et al., 2020; Liu et al., 2024; Ramos et al., 2022; Rauf et al., 2025) Their distribution across the main compound classes is summarized in Figure S7. Several metabolites associated with the phenylpropanoid pathway and fruit antioxidant capacity were identified, including chlorogenic acid, neochlorogenic acid, catechin, naringenin, trifolin, kuromanin, and 4-hydroxy-3-(3-methylbut-2-en-1-yl)benzoic acid. Primary metabolites such as quinic acid, phenylalanine, tryptophan, adenine, and sorbitol were also detected, representing key intermediates in plant metabolism. In addition, several lipid-related compounds were annotated, including α-linolenic acid, linoleic acid (9,12-octadecadienoic acid), the oxylipins 13(S)-HOTrE and (15Z)-9,12,13-trihydroxy-15-octadecenoic acid, and the monoacylglycerol 1-stearoylglycerol, which are associated with membrane structure, oxidative signalling, and plant stress responses. Two triterpenoids were also identified, namely ursolic acid, a major constituent of fruit cuticular

waxes, and gypsogenin. Finally, the plant hormone abscisic acid, together with a structually related glycoside ((2R,3R,4S,5S,6R)-2-{[4-(4-hydroxy-2,6,6-trimethyl-cyclohex-1-en-1-yl)butan-2-yl]oxy}-6-(hydroxymethyl)oxane-3,4,5-triol), was identified both associated with fruit development and ripening. The detected metabolites are consistent with the known metabolomic composition of peach and nectarine fruits and provide biologically relevant information that complements pesticide residue profiling.

3.4 Dietary Risk Assessment Results

The results of the acute and chronic dietary risk assessment are summarized in Table 2. For all quantified pesticides, both the individual Hazard Quotients and the cumulative Hazard Index remained well below the threshold value of 1 in both the adult and the children exposure scenarios, indicating that the consumption of the analyzed peaches and nectarines is unlikely to pose an appreciable risk to consumer health. For chronic exposure, the cumulative Hazard Index was negligible in both population groups, more than two orders of magnitude below unity. Even the most frequently detected residues, namely boscalid, tebuconazole and cyprodinil, which dominated the occurrence pattern (Section 3.1), contributed only marginally to chronic exposure, confirming that their high detection frequency was not matched by toxicologically relevant intake levels. For acute exposure, which represents the more conservative worst-case scenario based on the highest residue concentration combined with a large-portion consumption event, the cumulative aHQ-based index reached 0.21 for adults and 0.56 for children. The higher value for the children scenario reflects their substantially greater intake-to-body-weight ratio, as captured by the critical toddler population used in the PRIMo model. Although these values are higher than the chronic estimates as expected from the use of the maximum rather than the mean residue level, they remained below unity, and therefore no acute health concern was identified. The acute risk profile was governed primarily by acetamiprid (aHQ = 0.30 in the children scenario), with a smaller contribution from

Journal Pre-proof

tebuconazole; their share arose from comparatively low acute reference doses rather than from particularly elevated residue concentrations (Table 2). This illustrates that, in cumulative acute assessments, compounds with stringent toxicological reference values can dominate the risk profile even when present at low levels.

[Table 2]

The distribution of the cumulative chronic risk across the individual samples is shown in Figure S8 for both scenarios. When the Hazard Index was computed on a per-sample basis (Table S12), rather than from the dataset-wide mean, every one of the 74 samples remained well below the threshold of concern in both scenarios. The most exposed individual sample (sample 39) reached a Hazard Index of only 2.3×10^{-2} (adults) and 2.1×10^{-2} (children), approximately 45-fold below unity. This per-sample analysis confirms that the conclusion of acceptable dietary risk holds not merely on average but for each individual fruit sample analyzed, including the most heavily contaminated ones. A more detailed picture of which compounds drive the cumulative risk in each sample is provided in Figure S9, which presents the chronic Hazard Quotient contribution of every pesticide across the 74 samples. Emamectin benzoate emerged as a prominent contributor in several samples despite its consistently low concentrations which is a direct consequence of its exceptionally stringent ADI, whereas frequently detected compounds such as fludioxonil and cyprodinil contributed only faintly, reflecting their high toxicological reference values. The two panels share the same contribution pattern, since the difference between scenarios enters solely through the common intake-to-body-weight factor. This decoupling between contamination load and toxicological relevance is further illustrated in Figure S10, which breaks down the cumulative chronic Hazard Index into its compound-specific contributions for the samples of greatest relevance. The figure makes clear that the cumulative risk of a given sample is governed by its compound composition rather than by its total residue burden: samples with a comparatively low overall

residue load but containing residues of low-ADI substances (e.g. emamectin benzoate, tetraconazole) attained higher Hazard Index values than samples with a substantially greater total residue mass dominated by tolerant-ADI compounds. This finding reinforces the principle that, in dietary risk assessment, the identity and toxicological potency of the residues present are more decisive than the absolute quantity of pesticide detected. It should be noted that the cumulative Hazard Index used here was based on simple dose addition of the individual HQs, irrespective of differences in the pesticides' modes of action, and should therefore be regarded as a conservative screening-level estimate of cumulative risk.

4. Conclusions

Herein, an integrated LC-HRMS workflow combining target, suspect, and non-target screening was applied to peach and nectarine samples from major Greek production areas. The results demonstrate that target analysis alone provides only part of the available chemical information. While it supported regulatory assessment of pesticide residues, suspect screening extended the investigation to transformation products of confirmed parent pesticides, and non-target screening further broadened sample characterization by revealing food-contact/material- and packaging-related emerging contaminants and endogenous compounds. Notably, no additional high-confidence pesticide residues were identified by non-target screening beyond those already confirmed by target analysis. Dietary risk assessment indicated that acute, chronic, and cumulative exposure remained well below the corresponding health-based guidance values, suggesting no appreciable consumer risk under the evaluated exposure scenarios. Importantly, the combined findings support the initial premise that integrated LC-HRMS analysis provides broader chemical characterization than target analysis alone and can better capture the complexity of pesticide-related occurrence in food matrices.

Finally, the proposed workflow highlights the value of full-scan HRMS data for simultaneously supporting regulatory residue monitoring, transformation-product

investigation, broader chemical profiling, and dietary risk assessment. Such integrated strategies may strengthen future food-monitoring programs by maximizing the information obtained from a single analytical dataset and by enabling more comprehensive evaluation of both regulated residues and additional pesticide-related or food-associated chemical signals. Further improvement in transformation-product identification would benefit from the availability of additional reference standards and expanded high-resolution spectral libraries.

Funding

This research was funded by European Commission under the HORIZON-WIDERA-TWINNING BOTTOM-UP within the framework of the SPECTRA project (Grant No 101158453).

CRediT author statement

Vasileios Alampanos: Formal analysis, Investigation, Methodology, Validation, Data curation, Visualization, Supervision, Conceptualization, Software, Writing - Original draft; **Lelouda-Athanasia Koronaiou:** Formal analysis, Investigation, Methodology, Data curation, Validation, Visualization, Software, Writing - Review & Editing; **Styliani Petromelidou**: Methodology, Investigation, Validation; **Anthia D. Gkantou:** Methodology, Investigation, Validation; **Theocharis G. Danis:** Methodology, Investigation, Validation; **Dimitra A. Lambropoulou:** Investigation, Methodology, Validation, Conceptualization, Writing - Review & Editing, Supervision, Funding acquisition, Resources, Project administration.

References

Åkesson, A., Donat-Vargas, C., Hallström, E., Sonesson, U., Widenfalk, A., & Wolk, A. (2023). Associations between dietary pesticide residue mixture exposure and mortality in a

population-based prospective cohort of men and women. *Environment International*, *182*, 108346. https://doi.org/10.1016/j.envint.2023.108346

Alampanos, V. D., & Lambropoulou, D. A. (2025). Identification of metabolites and transformation products of environmental contaminants. In *Sample Handling and Trace Analysis of Pollutants* (pp. 663–690). Elsevier Science. https://doi.org/10.1016/B978-0-323-85601-0.00023-0

Anagnostopoulpou, K., Nannou, C., Aschonitis, V. G., & Lambropoulou, D. A. (2022). Screening of pesticides and emerging contaminants in eighteen Greek lakes by using target and non-target HRMS approaches: Occurrence and ecological risk assessment. *Science of The Total Environment*, *849*, 157887. https://doi.org/10.1016/j.scitotenv.2022.157887

Apell, J. N., Pflug, N. C., & McNeill, K. (2019). Photodegradation of Fludioxonil and Other Pyrroles: The Importance of Indirect Photodegradation for Understanding Environmental Fate and Photoproduct Formation. *Environmental Science & Technology*, *53*(19), 11240–11250. https://doi.org/10.1021/acs.est.9b03948

Bakırcı, G. T., Yaman Acay, D. B., Bakırcı, F., & Ötleş, S. (2014). Pesticide residues in fruits and vegetables from the Aegean region, Turkey. *Food Chemistry*, *160*, 379–392. https://doi.org/10.1016/j.foodchem.2014.02.051

Bonini, P., Kind, T., Tsugawa, H., Barupal, D. K., & Fiehn, O. (2020). Retip: Retention Time Prediction for Compound Annotation in Untargeted Metabolomics. *Analytical Chemistry*, *92*(11), 7515–7522. https://doi.org/10.1021/acs.analchem.9b05765

Cao, S., Du, H., Tang, B., Xi, C., & Chen, Z. (2021). Non-target metabolomics based on high-resolution mass spectrometry combined with chemometric analysis for discriminating geographical origins of *Rhizoma Coptidis*. *Microchemical Journal*, *160*, 105685. https://doi.org/10.1016/j.microc.2020.105685

Casals, C., Guijarro, B., De Cal, A., Torres, R., Usall, J., Perdrix, V., Hilscher, U., Ladurner, E., Smets, T., & Teixidó, N. (2021). Field validation of biocontrol strategies to control brown rot on stone fruit in several European countries. *Pest Management Science*, *77*(5), 2502–2511. https://doi.org/10.1002/ps.6281

Journal Pre-proof

Chen, T., Zhang, Y., Wang, Y., Liang, W., Yan, Z., Lu, X., Liu, X., Zhao, C., & Xu, G. (2025). Suspect and nontarget screening of pesticides and their transformation products in agricultural products using liquid chromatography–high-resolution mass spectrometry. *Talanta*, *283*, 127154. https://doi.org/10.1016/j.talanta.2024.127154

Covarrubias, M. P., Lillo-Carmona, V., Melet, L., Benedetto, G., Andrade, D., Maucourt, M., Deborde, C., Fuentealba, C., Moing, A., Valenzuela, M. L., Pedreschi, R., & Almeida, A. M. (2020). Metabolite Fruit Profile Is Altered in Response to Source-Sink Imbalance and Can Be Used as an Early Predictor of Fruit Quality in Nectarine. *Frontiers in Plant Science*, *11*, 604133. https://doi.org/10.3389/fpls.2020.604133

Cui, K., Fang, L., Ding, R., Ni, R., Guan, S., Liang, J., Li, T., Liu, J., Wang, J., Dong, Z., Wu, X., & Zheng, Y. (2025). Dissipation, metabolism, processing factors and risk assessment of pesticides in peaches from field cultivation to crisp processing. *Food Chemistry: X*, *30*, 102974. https://doi.org/10.1016/j.fochx.2025.102974

EFSA. (2010). Management of left-censored data in dietary exposure assessment of chemical substances. *EFSA Journal*, *8*(3), 1557. https://doi.org/10.2903/j.efsa.2010.1557

EFSA. (2019). *PRIMo rev 3.1—Pesticide Residue Intake Model calculator | Zenodo*. https://zenodo.org/records/2620033

EFSA. (2026, February 10). *Pesticide active substances | EFSA*. https://www.efsa.europa.eu/en/topics/pesticide-active-substances

EFSA, Cabrera, L. C., Di Piazza, G., Dujardin, B., Marchese, E., & Medina Pastor, P. (2025). The 2023 European Union report on pesticide residues in food. *EFSA Journal*, *23*(5), e9398. https://doi.org/10.2903/j.efsa.2025.9398

EFSA, Cabrera, L. C., Dujardin, B., Marchese, E., & Medina Pastor, P. (2024). The 2022 European Union report on pesticide residues in food. *EFSA Journal*, *22*(4), e8753. https://doi.org/10.2903/j.efsa.2024.8753

EFSA, Medina Pastor, P., Carrasco Cabrera, L., Di Piazza, G., & González Ciria, C. (2026). The 2024 European Union report on pesticide residues in food. *EFSA Journal*, *24*(5), e10054. https://doi.org/10.2903/j.efsa.2026.10054

Enterprise Greece. (2026). *Peaches*. https://www.enterprisegreece.gov.gr/en/trade/key-sectors/food-beverage/peaches/

European Commission. (n.d.). *EU Pesticides Database—MRLs—Search Pesticide Residues*. EU Pesticides Database. Retrieved February 28, 2026, from https://ec.europa.eu/food/plant/pesticides/eu-pesticides-database/start/screen/mrls/searchpr

European Commission. (2016, October 14). *Commission Implementing Regulation (EU) 2016/1826*. European Union. http://data.europa.eu/eli/reg_impl/2016/1826/oj

European Commission, Directorate-General for Health and Food Safety. (2026). *Analytical quality control and method validation procedures for pesticide residues analysis in Food and Feed—SANTE 11312/2021 v2026*.

Eyring, P., Tienstra, M., Mol, H., Herrmann, S. S., Rasmussen, P. H., Frandsen, H. L., & Poulsen, M. E. (2021). Development of a new generic extraction method for the analysis of pesticides, mycotoxins, and polycyclic aromatic hydrocarbons in representative animal feed and food samples. *Food Chemistry*, *356*, 129653. https://doi.org/10.1016/j.foodchem.2021.129653

FAOSTAT. (2023). FAO. https://www.fao.org/faostat/en/#data/QCL/visualize

FAO/WHO. (2026). *JMPR - FAO/WHO Joint Meeting on Pesticide Residues | Pest and Pesticide Management | Food and Agriculture Organization of the United Nations | IPM and Pesticide Risk Reduction | Food and Agriculture Organization of the United Nations*. https://www.fao.org/pest-and-pesticide-management/guidelines-standards/faowho-joint-meeting-on-pesticide-residues-jmpr/en/

FAO/WHO Codex Alimentarius Commission. (n.d.). *Codex Committee on Pesticide Residues (CCPR)*. Codex Alimentarius. Retrieved February 24, 2026, from https://www.fao.org/fao-who-codexalimentarius/committees/committee/en/?committee=CCPR

Fenik, J., Tankiewicz, M., & Biziuk, M. (2011). Properties and determination of pesticides in fruits and vegetables. *TrAC Trends in Analytical Chemistry*, *30*(6), 814–826. https://doi.org/10.1016/j.trac.2011.02.008

García-Reyes, J. F., Molina-Díaz, A., & Fernandez-Alba, A. R. (2007). Identification of pesticide transformation products in food by liquid chromatography/time-of-flight mass spectrometry

Journal Pre-proof

via “fragmentation-degradation” relationships. *Analytical Chemistry*, *79*(1), 307–321. https://doi.org/10.1021/ac061402d

Gkountouras, D., Boti, V., & Albanis, T. (2024). High resolution mass spectrometry targeted analysis and suspect screening of pesticide residues in fruit samples and assessment of dietary exposure. *Environmental Pollution*, *352*, 124143. https://doi.org/10.1016/j.envpol.2024.124143

Grapholita molesta—An overview | ScienceDirect Topics. (n.d.). Retrieved June 6, 2026, from https://www.sciencedirect.com/topics/agricultural-and-biological-sciences/grapholita-molesta

Hernández, F., Cervera, M. I., Portolés, T., Beltrán, J., & Pitarch, E. (2013). The role of GC-MS/MS with triple quadrupole in pesticide residue analysis in food and the environment. *Analytical Methods*, *5*(21), 5875–5894. https://doi.org/10.1039/C3AY41104D

Jankowska, M., Kaczyński, P., Hrynko, I., Rutkowska, E., Iwaniuk, P., Ilyasova, G., & Łozowicka, B. (2024). Dietary risk assessment of children and adults consuming fruit and vegetables with multiple pesticide residues. *Chemosphere*, *369*, 143858. https://doi.org/10.1016/j.chemosphere.2024.143858

Kang, D., Doudrick, K., Park, N., Choi, Y., Kim, K., & Jeon, J. (2020). Identification of transformation products to characterize the ability of a natural wetland to degrade synthetic organic pollutants. *Water Research*, *187*, 116425. https://doi.org/10.1016/j.watres.2020.116425

Kaufmann, A. (2012). The current role of high-resolution mass spectrometry in food analysis. *Analytical and Bioanalytical Chemistry*, *403*(5), 1233–1249. https://doi.org/10.1007/s00216-011-5629-4

Keklik, M., Golge, O., González-Curbelo, M. Á., & Kabak, B. (2025). Pesticide residues in peaches and nectarines: Three-year monitoring data and risk assessment. *Food Control*, *171*, 111141. https://doi.org/10.1016/j.foodcont.2025.111141

Kiefer, K., Müller, A., Singer, H., & Hollender, J. (2019). New relevant pesticide transformation products in groundwater detected using target and suspect screening for agricultural and urban

micropollutants with LC-HRMS. *Water Research*, *165*, 114972. https://doi.org/10.1016/j.watres.2019.114972

Kottadiyil, D., Mehta, T., Thasale, R., & P, S. (2023). Determination and dietary risk assessment of 52 pesticide residues in vegetable and fruit samples by GC-MS/MS and UHPLC-QTOF/MS from Gujarat, India. *Journal of Food Composition and Analysis*, *115*, 104957. https://doi.org/10.1016/j.jfca.2022.104957

Kunzelmann, M., Winter, M., Åberg, M., Hellenäs, K.-E., & Rosén, J. (2018). Non-targeted analysis of unexpected food contaminants using LC-HRMS. *Analytical and Bioanalytical Chemistry*, *410*(22), 5593–5602. https://doi.org/10.1007/s00216-018-1028-4

LCSB, U. of L.-. (n.d.). *FAIR-TPs—Transformation Products Browser*. Retrieved June 11, 2026, from https://fairtps.lcsb.uni.lu/

Li, C., Yuan, S., Xie, Y., Guo, Y., Cheng, Y., Yu, H., Qian, H., & Yao, W. (2021). Transformation of fluopyram during enzymatic hydrolysis of apple and its effect on polygalacturonase and apple juice yield. *Food Chemistry*, *357*, 129842. https://doi.org/10.1016/j.foodchem.2021.129842

Li, Z., Tan, M., Deng, H., Yang, X., Yu, Y., Zhou, D., & Dong, H. (2022). Geographical Origin Differentiation of Rice by LC–MS-Based Non-Targeted Metabolomics. *Foods*, *11*(21), 3318. https://doi.org/10.3390/foods11213318

Lillo-Carmona, V., Espinoza, A., Rothkegel, K., Rubilar, M., Nilo-Poyanco, R., Pedreschi, R., Campos-Vargas, R., & Meneses, C. (2020). Identification of Metabolite and Lipid Profiles in a Segregating Peach Population Associated with Mealiness in Prunus persica (L.) Batsch. *Metabolites*, *10*(4), 154. https://doi.org/10.3390/metabo10040154

Liu, Y., Wu, J., Li, Y., Deng, W., Cao, K., Li, Z., & Wang, L. (2024). Metabolism and transcriptional regulation in chilling injury development of nectarine fruit during postharvest cold storage. *Postharvest Biology and Technology*, *210*, 112748. https://doi.org/10.1016/j.postharvbio.2023.112748

Mahler, B. J., Nowell, L. H., Sandstrom, M. W., Bradley, P. M., Romanok, K. M., Konrad, C. P., & Van Metre, P. C. (2021). Inclusion of Pesticide Transformation Products Is Key to Estimating

Journal Pre-proof

Pesticide Exposures and Effects in Small U.S. Streams. *Environmental Science & Technology*, *55*(8), 4740–4752. https://doi.org/10.1021/acs.est.0c06625

Man, Y., Stenrød, M., Wu, C., Almvik, M., Holten, R., Clarke, J. L., Yuan, S., Wu, X., Xu, J., Dong, F., Zheng, Y., & Liu, X. (2021). Degradation of difenoconazole in water and soil: Kinetics, degradation pathways, transformation products identification and ecotoxicity assessment. *Journal of Hazardous Materials*, *418*, 126303. https://doi.org/10.1016/j.jhazmat.2021.126303

Marion Murray et al. (2020). *The Backyard Orchardist: Fruit Pests: Peach & Nectarine*. https://extension.usu.edu/planthealth/research/backyard-peach-nectarine-pests

Mayahi, B., Palm, E. H., & Schymanski, E. L. (2026). Browsing FAIR Transformation Product Information with FAIR-TPs. *ACS Environmental Au*, *6*(3), 513–522. https://doi.org/10.1021/acsenvironau.5c00314

Mitra, S., Saran, R. K., Srivastava, S., & Rensing, C. (2024). Pesticides in the environment: Degradation routes, pesticide transformation products and ecotoxicological considerations. *Science of The Total Environment*, *935*, 173026. https://doi.org/10.1016/j.scitotenv.2024.173026

Mohammed Taha, H., Aalizadeh, R., Alygizakis, N., Antignac, J.-P., Arp, H. P. H., Bade, R., Baker, N., Belova, L., Bijlsma, L., Bolton, E. E., Brack, W., Celma, A., Chen, W.-L., Cheng, T., Chirsir, P., Čirka, Ľ., D'Agostino, L. A., Djoumbou Feunang, Y., Dulio, V., … Schymanski, E. L. (2022). The NORMAN Suspect List Exchange (NORMAN-SLE): Facilitating European and worldwide collaboration on suspect screening in high resolution mass spectrometry. *Environmental Sciences Europe*, *34*(1), 104. https://doi.org/10.1186/s12302-022-00680-6

Pesticides | EFSA. (2026, February 11). EFSA. https://www.efsa.europa.eu/en/topics/topic/pesticides

PubChem. (n.d.). *PubChem*. Retrieved June 11, 2026, from https://pubchem.ncbi.nlm.nih.gov/

Ramos, M., Ghosson, H., Raviglione, D., Bertrand, C., & Salvia, M.-V. (2022). Untargeted metabolomics as a tool to monitor biocontrol product residues' fate on field-treated Prunus persica. *The Science of the Total Environment*, *807*(Pt 1), 150717. https://doi.org/10.1016/j.scitotenv.2021.150717

Journal Pre-proof

Rauf, A., Olatunde, A., Akram, Z., Hemeg, H. A., Khalil, A. A., Shariati, M. A., Edis, Z., Thiruvengadam, M., & Kim, S.-H. (2025). Metabolomic-based promising technique for fruit-based natural products analysis. *Food Chemistry: X*, *32*, 103374. https://doi.org/10.1016/j.fochx.2025.103374

ReportLinker. (2023). *European Peaches and Nectarines Harvested Production Share by Country (Thousand Metric Tons), 2023*. https://www.reportlinker.com/dataset/fb889f476233c6b53b7bb0dfc6c63e202496ca27

Romero, A., Ramos, E., Ares, I., Castellano, V., Martínez, M., Martínez-Larrañaga, M. R., Anadón, A., & Martínez, M. A. (2016). Fipronil sulfone induced higher cytotoxicity than fipronil in SH-SY5Y cells: Protection by antioxidants. *Toxicology Letters*, *252*, 42–49. https://doi.org/10.1016/j.toxlet.2016.04.005

Sandoval-Insausti, H., Chiu, Y.-H., Wang, Y.-X., Hart, J. E., Bhupathiraju, S. N., Mínguez-Alarcón, L., Ding, M., Willett, W. C., Laden, F., & Chavarro, J. E. (2022). Intake of fruits and vegetables according to pesticide residue status in relation to all-cause and disease-specific mortality: Results from three prospective cohort studies. *Environment International*, *159*, 107024. https://doi.org/10.1016/j.envint.2021.107024

Schusterova, D., Han, J., Gomersall, V., Jursik, M., Horska, T., Kosek, V., Kocourek, F., Kocourek, V., & Hajslova, J. (2025). Optimized methods for the investigation of changes in levels of pesticide residues and their transformation products in iceberg lettuce. *Food Research International*, *202*, 115625. https://doi.org/10.1016/j.foodres.2024.115625

Schymanski, E. L., Jeon, J., Gulde, R., Fenner, K., Ruff, M., Singer, H. P., & Hollender, J. (2014). Identifying Small Molecules via High Resolution Mass Spectrometry: Communicating Confidence. *Environmental Science & Technology*, *48*(4), 2097–2098. https://doi.org/10.1021/es5002105

Si, W., Song, Y., Guo, C., Huang, Z., Chen, L., Xu, F., Bai, B., Huo, K., & Wang, S. (2025). Multi-model risk assessment of pesticide residues in seasonal fruits based on integrated targeted/non-targeted screening: Implications for vulnerable populations. *Food Chemistry: X*, *28*, 102599. https://doi.org/10.1016/j.fochx.2025.102599

Journal Pre-proof

Słowik-Borowiec, M., Szpyrka, E., Rupar, J., Matyaszek, A., & Podbielska, M. (2015). Pesticide residues in stone fruits from the south-eastern region of Poland in 2012-2014. *Roczniki Państwowego Zakładu Higieny*, *66*(3), 211–216.

Statistics—ELSTAT. (n.d.). Retrieved February 28, 2026, from https://www.statistics.gr/en/statistics/-/publication/SPG06/-

Tamasi, G., Bonechi, C., Leone, G., Andreassi, M., Consumi, M., Sangiorgio, P., Verardi, A., Rossi, C., & Magnani, A. (2021). Varietal and Geographical Origin Characterization of Peaches and Nectarines by Combining Analytical Techniques and Statistical Approach. *Molecules*, *26*(14), 4128. https://doi.org/10.3390/molecules26144128

Thomidis, T., & Michailides, T. J. (2009). Studies on *Diaporthe eres* as a New Pathogen of Peach Trees in Greece. *Plant Disease*, *93*(12), 1293–1297. https://doi.org/10.1094/PDIS-93-12-1293

Tudi, M., Daniel Ruan, H., Wang, L., Lyu, J., Sadler, R., Connell, D., Chu, C., & Phung, D. T. (2021). Agriculture Development, Pesticide Application and Its Impact on the Environment. *International Journal of Environmental Research and Public Health*, *18*(3), 1112. https://doi.org/10.3390/ijerph18031112

Ueno, E., Oshima, H., Matsumoto, H., Saito, I., & Tamura, H. (2006). Determination of spinosad in vegetables and fruits by high-performance liquid chromatography with UV and mass spectrometric detection after gel permeation chromatography and solid-phase extraction cleanup on a 2-layered column. *Journal of AOAC International*, *89*(6), 1641–1649.

US EPA, O. (2015, January 14). *About Pesticide Tolerances* [Overviews and Factsheets]. https://www.epa.gov/pesticide-tolerances/about-pesticide-tolerances

USDA Foreign Agricultural Service. (2026). *Peaches & Nectarines*. https://www.fas.usda.gov/data/production/0579309

Wang, F., Allen, D., Tian, S., Oler, E., Gautam, V., Greiner, R., Metz, T. O., & Wishart, D. S. (2022). CFM-ID 4.0—A web server for accurate MS-based metabolite identification. *Nucleic Acids Research*, *50*(W1), W165–W174. https://doi.org/10.1093/nar/gkac383

World Health Organization. (n.d.). *Global Environment Monitoring System (GEMS) / Food Contamination Monitoring and Assessment Programme*. Retrieved June 25, 2026, from

https://www.who.int/teams/nutrition-and-food-safety/databases/global-environment-monitoring-system-food-contamination

World Population Review. (2026, September 10). *Peach Production by Country 2026*. https://worldpopulationreview.com/country-rankings/peach-production-by-country

Yang, J., Wang, Z., Lv, G., Liu, W., Wang, Y., Sun, X., & Gao, J. (2020). Indirect photodegradation of fludioxonil by hydroxyl radical and singlet oxygen in aquatic environment: Mechanism, photoproducts formation and eco-toxicity assessment. *Ecotoxicology and Environmental Safety*, *197*, 110644. https://doi.org/10.1016/j.ecoenv.2020.110644

Yang, L., Shen, Q., Zeng, T., Li, J., Li, W., & Wang, Y. (2020). Enrichment of imidacloprid and its metabolites in lizards and its toxic effects on gonads. *Environmental Pollution*, *258*, 113748. https://doi.org/10.1016/j.envpol.2019.113748

Yang, Y., Zhang, Q., Xiao, Y., Xiao, Y., Gao, H., Zhang, S., Covaci, A., & Xia, X. (2025). Urban sewage discharge of neonicotinoids and their transformation products threatens aquatic organisms. *Water Research*, *268*, 122740. https://doi.org/10.1016/j.watres.2024.122740

Zhao, J., Li, A., Jin, X., Liang, G., & Pan, L. (2022). Discrimination of Geographical Origin of Agricultural Products From Small-Scale Districts by Widely Targeted Metabolomics With a Case Study on Pinggu Peach. *Frontiers in Nutrition*, *9*. https://doi.org/10.3389/fnut.2022.891302

Journal Pre-proof

Figures

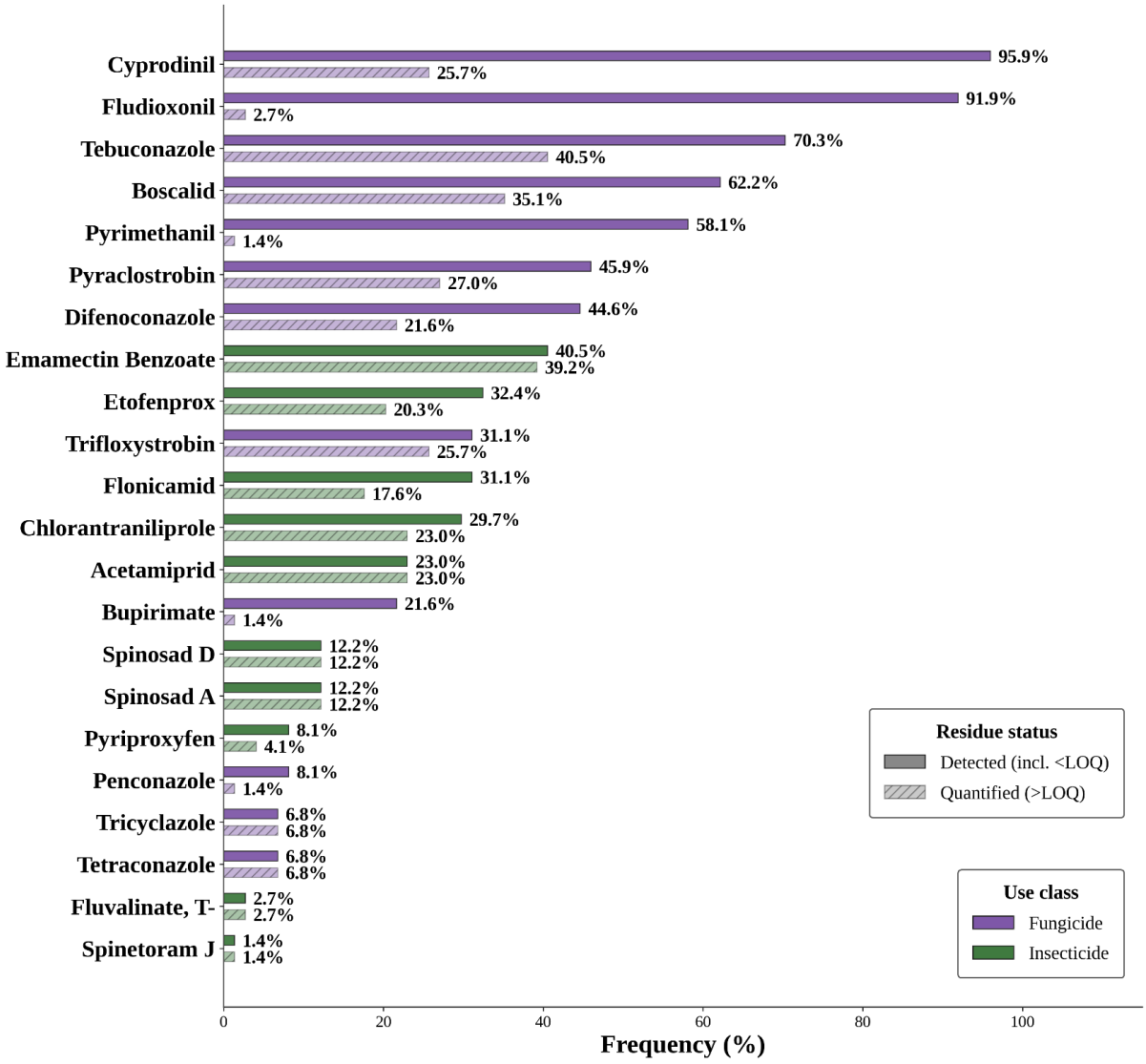


Figure 1 Detection and quantification frequencies of pesticide residues in peach and nectarine samples (n = 74) determined by LC-HRMS target screening.

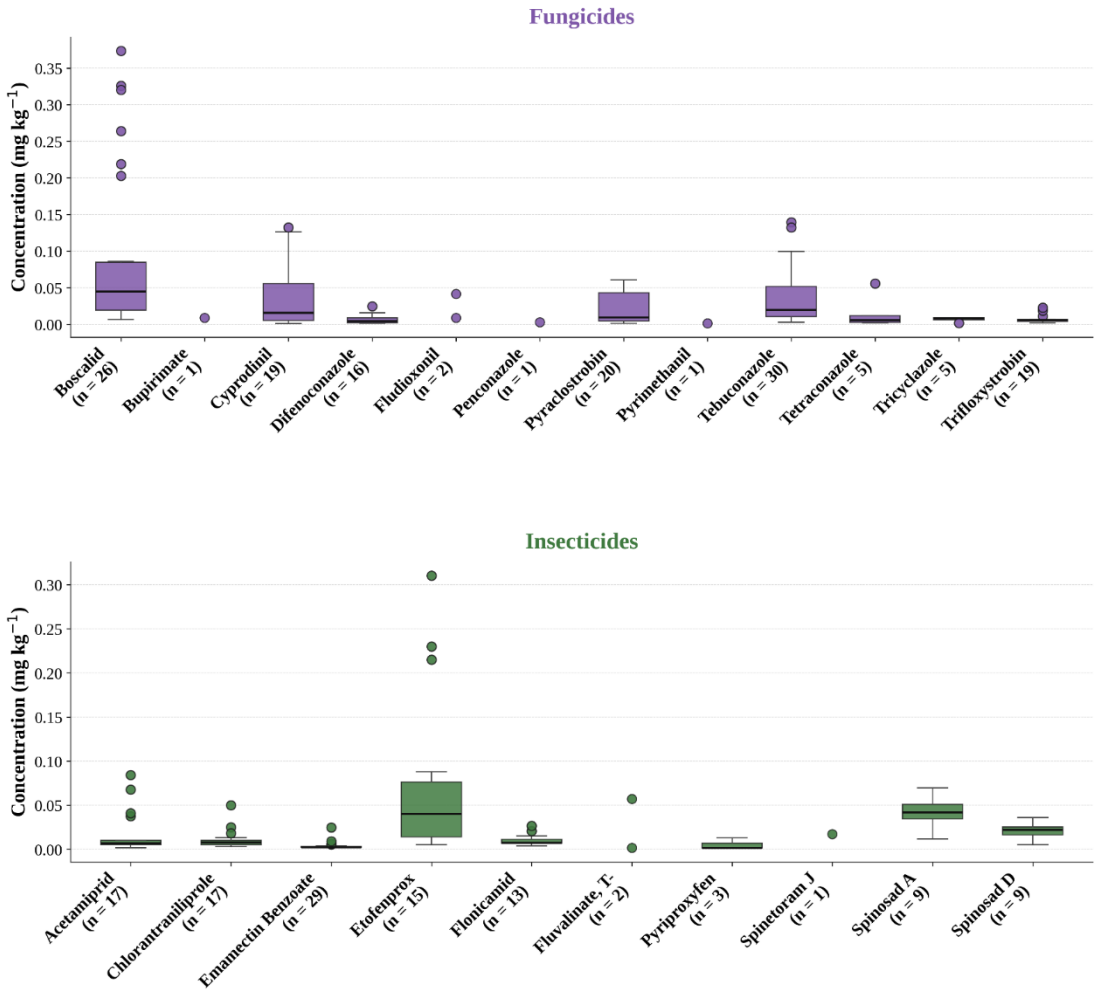


Figure 2 Boxplots of quantified pesticide residue concentrations (mg kg⁻¹) detected in peach and nectarine samples. Fungicides and insecticides are shown separately; n denotes the number of quantified detections per compound.

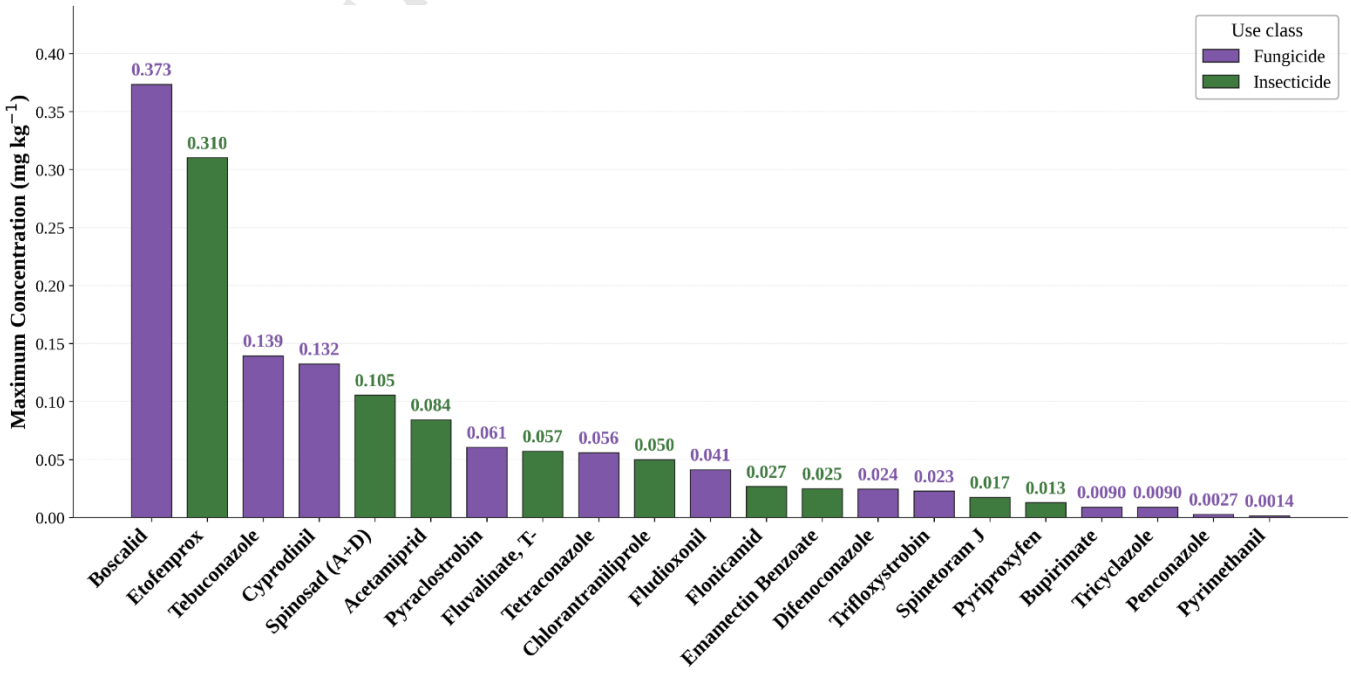


Figure 3 Maximum concentrations (mg kg $^{-1}$) of pesticide residues detected in peach and nectarine samples.

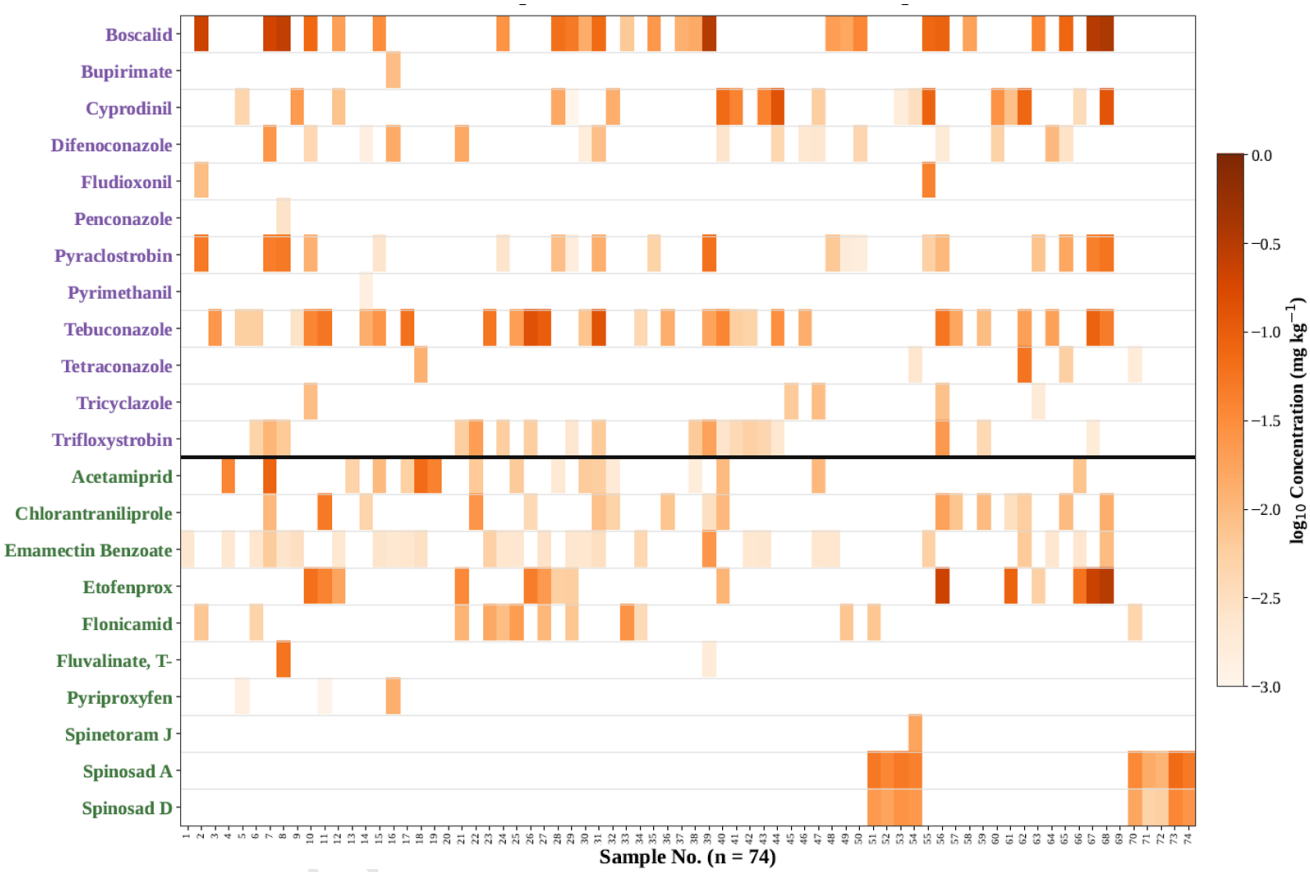


Figure 4 Heatmap illustrates the occurrence and concentration profiles of quantified fungicide and insecticide residues across peach and nectarine samples (n = 74).

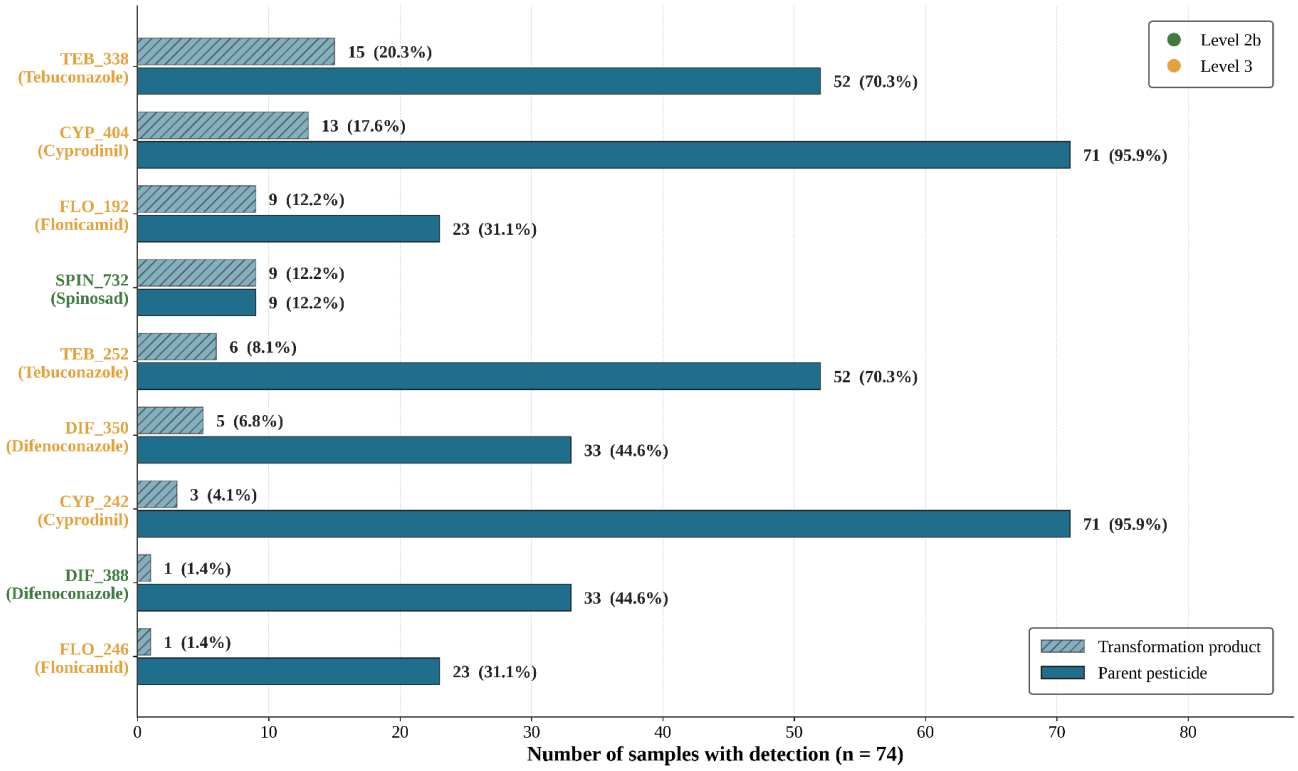


Figure 5 Detection frequencies of identified TPs and their corresponding parent pesticides.

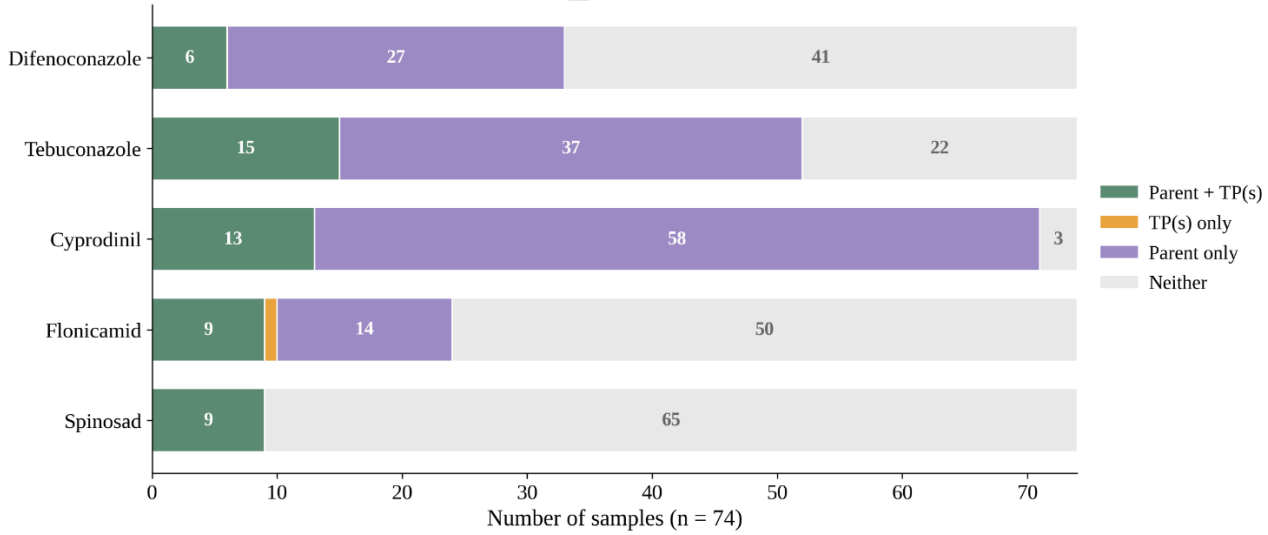


Figure 6 Co-occurrence patterns of parent pesticides and their TPs in samples (n = 74).

Tables

Table 1: Comparison of quantified pesticide residues detected in peach samples with the corresponding EU maximum residue limits (MRLs).

Active Substance	MRL (mg/kg)	Maximum Found Concentration (mg/kg)	Comparison with MRL	Number of Samples Above MRL
Acetamiprid	0.080	0.0843	**ABOVE MRL***	**1**
Boscalid	5.000	0.3734	**COMPLIANT**	-
Bupirimate	0.300	0.0090	**COMPLIANT**	-
Chlorantraniliprole	1.000	0.0500	**COMPLIANT**	-
Cyprodinil	2.000	0.1324	**COMPLIANT**	-
Difenoconazole	0.500	0.0245	**COMPLIANT**	-
Emamectin Benzoate	0.030	0.0248	**COMPLIANT**	-
Etofenprox	0.600	0.3102	**COMPLIANT**	-
Flonicamid	0.400	0.0267	**COMPLIANT**	-
Fludioxonil	10.000	0.0413	**COMPLIANT**	-
Fluvalinate, tau-	0.300	0.0572	**COMPLIANT**	-
Penconazole	0.070	0.0027	**COMPLIANT**	-
Pyraclostrobin	0.300	0.0605	**COMPLIANT**	-
Pyrimethanil	10.000	0.0014	**COMPLIANT**	-
Pyriproxyfen	0.500	0.0128	**COMPLIANT**	-
Spinetoram	0.300	0.0174	**COMPLIANT**	-
Spinosad A		0.0696	**See sum A+D**	-
Spinosad D		0.0359	**See sum A+D**	-
Spinosad (sum A+D)	0.600	0.1055	**COMPLIANT**	-
Tebuconazole	0.600	0.1393	**COMPLIANT**	-
Tetraconazole	0.030	0.0558	**ABOVE MRL***	**1**
Tricyclazole	0.010	0.0090	**COMPLIANT**	-
Trifloxystrobin	3.000	0.0227	**COMPLIANT**	-

**Measured concentrations above the numerical MRL should not considered as confirmed regulatory non-compliances because analyte-specific expanded measurement uncertainty was not taken into consideration.*

Table 2: Acute and chronic dietary risk assessment of the pesticide residues quantified in peach and nectarine samples (n = 74): individual Hazard Quotients per compound and the cumulative Hazard Index, for the adult (general-population) and children scenarios.

Pesticide	HR (mg kg⁻¹)	Mean (mg kg⁻¹)	ADI (mg kg⁻¹ bw)	ARfD (mg kg⁻¹ bw)	cHQ adults	cHQ children	aHQ adults	aHQ children
Acetamiprid	0.0843	0.0042	0.0050	0.0050	3.13E-04	2.90E-04	1.14E-01	3.03E-01
Boscalid	0.3734	0.0339	0.0400	n.r.	3.18E-04	2.95E-04	–	–
Bupirimate	0.0090	0.0004	0.0500	n.r.	3.20E-06	2.97E-06	–	–
Chlorantraniliprole	0.0500	0.0027	1.5600	n.r.	6.43E-07	5.96E-07	–	–
Cyprodinil	0.1324	0.0095	0.0300	n.r.	1.19E-04	1.10E-04	–	–
Difenoconazole	0.0245	0.0015	0.0100	0.1600	5.61E-05	5.20E-05	1.04E-03	2.75E-03
Emamectin benzoate	0.0248	0.0015	0.0005	0.0100	1.15E-03	1.07E-03	1.68E-02	4.46E-02
Etofenprox	0.3102	0.0157	0.0300	1.0000	1.97E-04	1.83E-04	2.11E-03	5.58E-03
Flonicamid	0.0267	0.0020	0.0250	0.0250	2.94E-05	2.73E-05	7.25E-03	1.92E-02
Fludioxonil	0.0413	0.0020	0.3700	n.r.	2.00E-06	1.85E-06	–	–
Fluvalinate, tau-	0.0572	0.0008	0.0050	0.0500	5.98E-05	5.55E-05	7.77E-03	2.06E-02
Penconazole	0.0027	0.0001	0.0300	0.5000	1.03E-06	9.53E-07	3.67E-05	9.72E-05
Pyraclostrobin	0.0605	0.0055	0.0300	0.0300	6.85E-05	6.35E-05	1.37E-02	3.63E-02
Pyrimethanil	0.0014	0.0002	0.0130	1.0000	6.29E-06	5.83E-06	9.50E-06	2.52E-05
Pyriproxyfen	0.0128	0.0002	0.0500	1.0000	1.64E-06	1.52E-06	8.69E-05	2.30E-04
Spinetoram J	0.0174	0.0002	0.0250	0.1000	3.54E-06	3.28E-06	1.18E-03	3.13E-03
Spinosad (A+D)	0.1055	0.0073	0.0240	0.1000	1.14E-04	1.05E-04	7.16E-03	1.90E-02
Tebuconazole	0.1393	0.0145	0.0300	0.0300	1.81E-04	1.68E-04	3.15E-02	8.36E-02
Tetraconazole	0.0558	0.0010	0.0040	0.0500	9.83E-05	9.11E-05	7.58E-03	2.01E-02
Tricyclazole	0.0090	0.0005	0.0500	0.0500	3.44E-06	3.19E-06	1.22E-03	3.24E-03
Trifloxystrobin	0.0227	0.0019	0.1000	0.5000	7.21E-06	6.68E-06	3.08E-04	8.17E-04
Cumulative chronic Hazard Index (HI = Σ cHQ)					**0.0027**	**0.0025**		
Cumulative acute Hazard Index (HI = Σ aHQ)							**0.2122**	**0.5627**
Conclusion					**< 1 (acceptable)**			

HR: highest residue; ADI: acceptable daily intake; ARfD: acute reference dose; n.r.: not required (no ARfD established — compound excluded from the acute assessment); cHQ/aHQ: chronic/acute Hazard Quotient; HI: Hazard Index. All HQ and HI values are unitless (threshold of concern = 1). Spinosad assessed as the sum of spinosyns A + D. Exposure inputs from EFSA PRIMo rev. 3.1; residues <LOQ substituted with LOQ/2 and non-detects with zero.

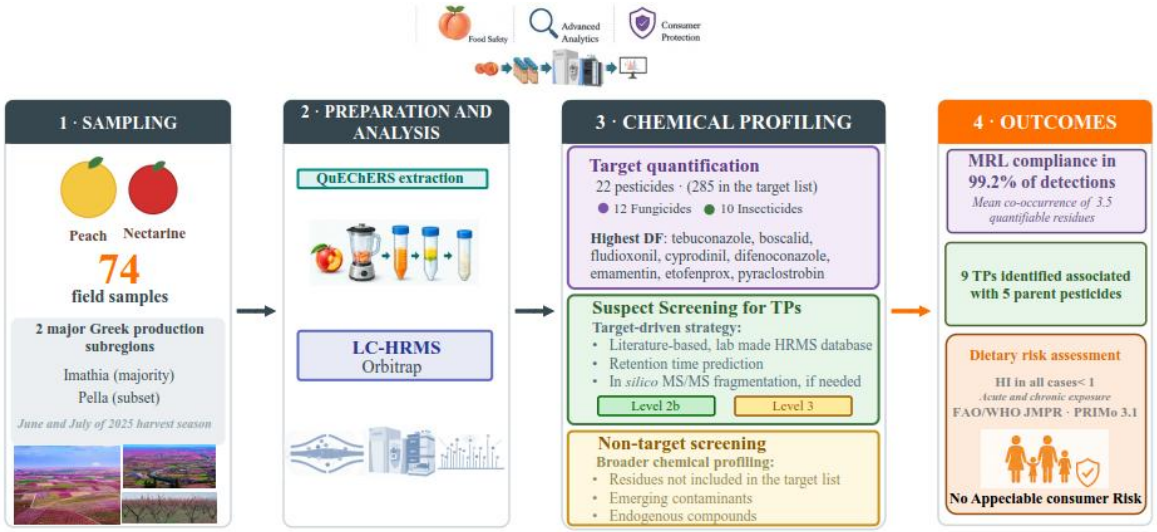
Declaration of interests

☒ The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

☐The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

Journal Pre-proof

Graphical abstract



Highlights

- LC-HRMS enabled comprehensive screening of 74 peach and nectarine samples.
- Target analysis identified 22 pesticide residues, 12 fungicides and 10 insecticides.
- 99.2% of residue detections were below the corresponding EU MRLs.
- Target-driven suspect screening revealed nine pesticide transformation products.
- Dietary risk assessment indicated no appreciable health risk for consumers.

Journal Pre-proof